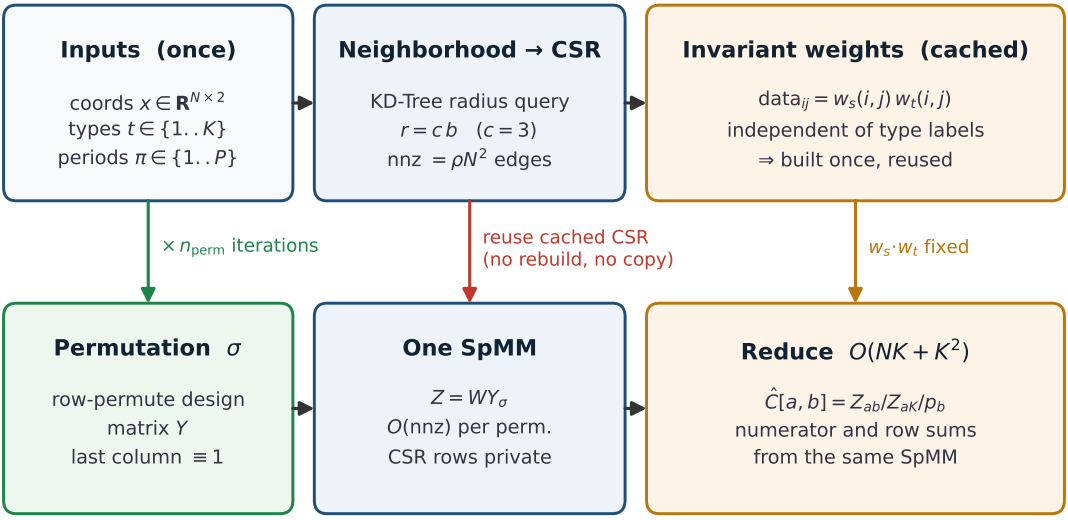
Structure-invariant sparse permutation inference: data flow
Inputs (once)
coords $x \in \mathbf{R}^{N\times 2}$
types $t \in \{1..K\}$
periods $\pi \in \{1..P\}$
Neighborhood → CSR
KD-Tree radius query
$r = c\,b \quad (c = 3)$
nnz $= \rho N^2$ edges
Invariant weights (cached)
$\text{data}_{ij} = w_s(i,j)\, w_t(i,j)$
independent of type labels
⇒ built once, reused
× n_{perm} iterations
reuse cached CSR
(no rebuild, no copy)
$w_s \cdot w_t$ fixed
Permutation σ
row-permute design
matrix Y
last column $\equiv 1$
One SpMM
$Z = WY_\sigma$
$O(\text{nnz})$ per perm.
CSR rows private
Reduce $O(NK + K^2)$
$\hat{C}[a,b] = Z_{ab}/Z_{aK}/p_b$
numerator and row sums
from the same SpMM